We divide this into four parts: $\mathcal{L}^\dagger \rho = W_1 + W_2 + W_3 + W_4$, where

$$W_1 = -(\nabla_z \rho) \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}),\ W_2 = -\rho \nabla_z \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}),$$
$$W_3 = -(\nabla_\xi \rho) \cdot \boldsymbol{g},\qquad\qquad W_4 = -\rho \nabla_\xi \cdot \boldsymbol{g}.$$

We seek perturbations of Hamiltonian dynamics with separable equilibria, i.e., such that $\rho = \rho_\beta(H(z))\hat{\rho}(\boldsymbol{\xi})$ is a stationary solution.

Taking this as an ansatz, the first term can be simplified as follows:

$$\begin{aligned} W_1 &= -\hat{\rho}(\boldsymbol{\xi})\rho'_\beta(H)\nabla_z H \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) \\ &= \beta\hat{\rho}(\boldsymbol{\xi})\rho_\beta(H)\nabla_z H \cdot \boldsymbol{\Phi} = \beta\rho\nabla_z H \cdot \boldsymbol{\Phi} \end{aligned}$$

as $\boldsymbol{J}$ is skew-symmetric.

Using the volume preservation of Hamiltonian systems, the second term satisfies

$$W_2 = -\rho\nabla_z \cdot (\boldsymbol{J}\nabla_z H + \boldsymbol{\Phi}) = -\rho\nabla_z \cdot \boldsymbol{\Phi},$$

hence

$$W_1 + W_2 = -\rho[\nabla_z - \beta\nabla_z H] \cdot \boldsymbol{\Phi}.$$

There are many possibilities for the distribution of $\boldsymbol{\xi}$, of course, but we will assume this is Gaussian for simplicity: $\hat{\rho}(\boldsymbol{\xi}) = \exp\left(-\beta \sum_{i=1}^m \mu_i \xi_i^2/2\right) = \exp(-\beta \boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{\xi}/2)$ to get

$$W_3 = -(\nabla_\xi \rho) \cdot \boldsymbol{g}(\boldsymbol{\xi}, z) = \beta\rho\boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{g}.$$

Finally, for an equilibrium solution we must have $W_1 + W_2 + W_3 + W_4 = 0$, so

$$-[\nabla_z - \beta\nabla_z H] \cdot \boldsymbol{\Phi} + \beta\boldsymbol{\xi}^T \boldsymbol{\mu}^{-1}\boldsymbol{g} - \nabla_\xi \cdot \boldsymbol{g} = 0.$$

This equation is the fundamental one for deterministic thermostats.

Example 8.1 To recover Nosé-Hoover, assume an even dimensional phase space, $z = (\boldsymbol{q}, \boldsymbol{p})$, $\mathrm{N_d}$ degrees of freedom, and $\xi \in \mathbb{R}$,

$$\boldsymbol{\Phi} = -\xi \begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{p} \end{bmatrix}, \qquad g(z) = \mu^{-1}(\boldsymbol{p} \cdot \nabla_{\boldsymbol{p}} H - \mathrm{N_d k_B T}).$$

Example 8.2 Slightly more generally, let

$$\boldsymbol{\Phi} = \xi \boldsymbol{G}(z), \qquad g(\xi, z) = g(z) = \mu^{-1}[\nabla_z H - \mathrm{k_B T}\nabla_z] \cdot \boldsymbol{G}(z).$$